

Ariski Myardi
1103220019
TK4604

TUGAS STM32

NUCLEO-F411RE

Board Filters

Commercial Part Number

Q

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PRODUCT INFO

Type

Supplier

MCU / MPU Series

Marketing Status

Price

MEMORY

FEATURES

Features

Large Picture

Docs & Resources

Datasheet

Buy

STM32F4 Series

NUCLEO-F446RE

STM32 Nucleo-64 development board with STM32F446RE MCU, supports Arduino and ST morpho connectivity

ACTIVE

Product is in mass production

Part Number : NUCLEO-F446RE
Commercial Part Number : NUCLEO-F446RE

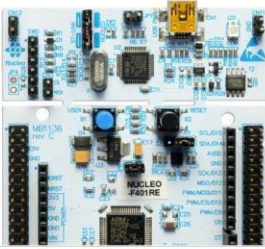
Unit Price (US\$) : 14.0
Mounted Device : STM32F446RETx

The STM32 Nucleo-64 board provides an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features provided by the STM32 microcontroller. For the compatible boards, the internal or external SMPS significantly reduces power consumption in Run mode.

The ARDUINO® Uno V3 connectivity support and the ST morpho headers allow the easy expansion of the functionality of the STM32 Nucleo open development platform with a wide choice of specialized shields.

The STM32 Nucleo-64 board does not require any separate probe as it integrates the ST-LINK debugger/programmer.

The STM32 Nucleo-64 board comes with the STM32 comprehensive free software libraries and examples available with the STM32Cube MCU Package.



Boards List: 2 items

	Overview	Commercial Part No	Type	Marketing Status	Unit Price (US\$)
☆		NUCLEO-F042K6	Nucleo-32	Active	10.32

NUCLEO-F412ZG

MCU/MPU Selector

Board Selector

Example Selector

Cross Selector

Board Filters

Commercial Part Number

Q

STM32F7

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PRODUCT INFO

Type

Supplier

MCU / MPU Series

Check/Uncheck All

STM32C0

STM32F0

STM32F1

STM32F2

STM32F3

STM32F4

STM32F7

STM32G0

STM32G4

STM32H5

STM32H7

STM32L0

STM32L1

STM32L4

STM32L4+

STM32L5

Aa [ab]

Features

Large Picture

Docs & Resources

Datasheet

Buy

STM32F4 Series

NUCLEO-F412ZG

STM32 Nucleo-144 development board with STM32F412ZG MCU, supports Arduino, ST Zio and morpho connectivity

ACTIVE

Product is in mass production

Part Number : NUCLEO-F412ZG
Commercial Part Number : NUCLEO-F412ZG

Unit Price (US\$) : 19.0
Mounted Device : STM32F412ZGTx

The STM32 Nucleo-144 board provides an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features, provided by the STM32 microcontroller. For the compatible boards, the internal or external SMPS significantly reduces power consumption in Run mode.

The ST Zio connector, which extends the ARDUINO® Uno V3 connectivity, and the ST morpho headers provide an easy means of expanding the functionality of the Nucleo open development platform with a wide choice of specialized shields.

The STM32 Nucleo-144 board does not require any separate probe as it integrates the ST-LINK debugger/programmer.

The STM32 Nucleo-144 board comes with the STM32 comprehensive free software libraries and examples available with the STM32Cube MCU Package.



Boards List: 201 items

	Overview	Commercial Part No	Type	Marketing Status	Unit Price (US\$)
☆		NUCLEO-F412ZG	Nucleo-144	Active	19.0

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TK4604
Perbandingan Kedua Board

Key Metrics	NUCLEO-F411RE	NUCLEO-F412ZG
Performance	Cortex-M4, 180 MHz, 512KB Flash, 128KB RAM	Cortex-M0, 100 MHz, 1MB Flash, 256KB RAM
Power Consumption	Low-power modes: Down to 7 μ A	Sedikit lebih banyak menggunakan daya di banding NUCLEO-F411RE
Peripheral Support	GPIO 50 pin, ADC: 1x12-bit, Timers: 11, I2C, USART, SPI, USB OTG	GPIO: 114 pin, Timers: 12, ADC: 1x 12-bit, I2C, USART, SPI, USB OTG, DFSDM

Contoh Aplikasi Embedded

Board: NUCLEO-F411RE

- **Aplikasi: Motor Control System:**
 - Menggunakan PWM untuk mengontrol kecepatan dan arah motor
 - Memanfaatkan ADC untuk umpan balik sensor posisi
 - Cocok untuk robotika atau otomasi industry

Board: NUCLEO-F411RE

- **Aplikasi: Pemrosesan Sinyal Digital:**
 - Menggunakan DFSDM untuk akuisisi sinyal audio berkualitas tinggi
 - Memanfaatkan DSP untuk filter dan analisis spektral
 - Aplikasi: pengenalan suara, analisis getaran

- **Referensi:**

<https://www.st.com/en/evaluation-tools/nucleo-f411re.html>

<https://www.st.com/en/evaluation-tools/nucleo-f412zg.html>